BEFORE YOU READ

Nine inverse pairs

Some functions are one-to-one as they stand; others need their domain restricted before an inverse exists.

L0 **Common inverse pairs**

L1 **Already one-to-one**

- L2 **Linear** requires m ≠ 0 — $mx + b \leftrightarrow (x - b)/m$
- L2 **Cubic** no restriction needed — $x^3 \leftrightarrow \sqrt[3]{x}$
- L2 **Exponential** $e^x \leftrightarrow \ln x$ as the special case — $a^x \leftrightarrow \log_a x$
- L2 **Reciprocal** self-inverse — $1/x \leftrightarrow 1/x$

L1 **Domain restriction required**

- L2 **Quadratic** on [0, ∞) — $x^2 \leftrightarrow \sqrt{x}$
- L2 **Power** even n needs the restriction, odd n does not — $x^n \leftrightarrow x\text{^}(1/n)$
- L2 **Sine** on [−π/2, π/2] — $\sin \leftrightarrow \arcsin$
- L2 **Cosine** on [0, π] — $\cos \leftrightarrow \arccos$
- L2 **Tangent** on (−π/2, π/2) — $\tan \leftrightarrow \arctan$

TAKEAWAY

Each pair is a reflection

Every pair in the list is a reflection across the line y = x.